Introduction to Sequences and Series

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Introduction to Sequences and Series

Sequences and series are fundamental concepts in calculus that deal with ordered lists of numbers and the summation of sequences, respectively. Understanding these concepts is crucial for analyzing functions, solving problems in mathematical analysis, and exploring applications in various fields such as physics, engineering, and economics.

Sequences

Definition

A **sequence** is an ordered list of numbers following a particular pattern. Each number in the sequence is called a **term**.

$$a_1, a_2, a_3, \ldots, a_n, \ldots$$

Types of Sequences

- **Arithmetic Sequence:** Each term is obtained by adding a constant difference to the preceding term.

  $$a_n = a_1 + (n-1)d$$

  where d is the common difference.

- **Geometric Sequence:** Each term is obtained by multiplying the preceding term by a constant ratio.

  $$a_n = a_1 \cdot r^{n-1}$$

  where r is the common ratio.

- **Harmonic Sequence:** The reciprocals of the terms form an arithmetic sequence.

  $$a_n = \frac{1}{n}$$

- **Fibonacci Sequence:** Each term is the sum of the two preceding terms.

  $$a_n = a_{n-1} + a_{n-2}$$

Convergence and Divergence

A sequence $\{a_n\}$ is said to **converge** to a limit L if for every $\epsilon > 0$, there exists an integer N such that for all $n > N$, $|a_n - L| < \epsilon$. If no such L exists, the sequence **diverges**.

Series

Definition

A **series** is the sum of the terms of a sequence.

$$S_n = a_1 + a_2 + a_3 + \cdots + a_n$$

Types of Series

- **Arithmetic Series:** The sum of an arithmetic sequence.

$$S_n = \frac{n}{2}(2a_1 + (n-1)d)$$

- **Geometric Series:** The sum of a geometric sequence.

$$S_n = a_1 \frac{1-r^n}{1-r} \quad \text{for } r \neq 1$$

- **Infinite Series:** A series with an infinite number of terms.

$$S = \sum_{n=1}^{\infty} a_n$$

Convergence and Divergence

An infinite series $\sum_{n=1}^{\infty} a_n$ **converges** to a limit S if the sequence of partial sums S_n approaches S as n approaches infinity. If S_n does not approach any finite limit, the series **diverges**.

Tests for Convergence

Several tests help determine whether an infinite series converges or diverges:

- **Nth-Term Test:** If $\lim_{n\to\infty} a_n \neq 0$, then $\sum a_n$ diverges.
- **Geometric Series Test:** A geometric series $\sum ar^{n-1}$ converges if $|r| < 1$ and diverges otherwise.
- **p-Series Test:** A p-series $\sum \frac{1}{n^p}$ converges if $p > 1$ and diverges if $p \leq 1$.
- **Integral Test:** If $a_n = f(n)$ where f is positive, continuous, and decreasing, then $\sum a_n$ and $\int_1^{\infty} f(x)\,dx$ both converge or both diverge.
- **Comparison Test:** Compare $\sum a_n$ with $\sum b_n$ where $0 \leq a_n \leq b_n$. If $\sum b_n$ converges, so does $\sum a_n$. If $\sum a_n$ diverges, so does $\sum b_n$.
- **Limit Comparison Test:** If $\lim_{n\to\infty} \frac{a_n}{b_n} = c$ where $0 < c < \infty$, then $\sum a_n$ and $\sum b_n$ either both converge or both diverge.
- **Ratio Test:** If $\lim_{n\to\infty} \left|\frac{a_{n+1}}{a_n}\right| = L$:
  - If $L < 1$, the series converges absolutely.
  - If $L > 1$ or $L = \infty$, the series diverges.
  - If $L = 1$, the test is inconclusive.
- **Root Test:** If $\lim_{n\to\infty} \sqrt[n]{|a_n|} = L$:
  - If $L < 1$, the series converges absolutely.

- If $L > 1$, the series diverges.
- If $L = 1$, the test is inconclusive.

- **Alternating Series Test:** For an alternating series $\sum(-1)^n a_n$:
  - If a_n is decreasing and $\lim_{n\to\infty} a_n = 0$, the series converges.

Examples

Example 1: Arithmetic Sequence

Find the 10th term of the arithmetic sequence where $a_1 = 3$ and $d = 2$.

Solution:

$$a_{10} = 3 + (10 - 1) \times 2 = 3 + 18 = 21$$

Example 2: Geometric Series

Determine whether the infinite geometric series $\sum_{n=1}^{\infty} 5\left(\frac{1}{3}\right)^{n-1}$ converges and find its sum if it does.

Solution: Since $|r| = \frac{1}{3} < 1$, the series converges.

$$S = \frac{a_1}{1 - r} = \frac{5}{1 - \frac{1}{3}} = \frac{5}{\frac{2}{3}} = \frac{15}{2} = 7.5$$

Example 3: p-Series Test

Determine the convergence of the series $\sum_{n=1}^{\infty} \frac{1}{n^2}$.

Solution: This is a p-series with $p = 2 > 1$, so the series converges.

Practice Problems

Sequences

1. Find the 15th term of the arithmetic sequence where $a_1 = -2$ and $d = 5$.
2. Determine the general formula for the geometric sequence with $a_1 = 4$ and $r = \frac{1}{2}$.
3. Is the sequence $a_n = \frac{3n+2}{n}$ convergent? If so, find its limit.
4. Find the first five terms of the Fibonacci sequence starting with $a_1 = 1$ and $a_2 = 1$.
5. Determine whether the sequence $a_n = (-1)^n$ converges or diverges.

Series

6. Determine whether the series $\sum_{n=1}^{\infty} 3\left(\frac{2}{5}\right)^n$ converges or diverges. If it converges, find its sum.
7. Use the p-Series Test to determine the convergence of $\sum_{n=1}^{\infty} \frac{1}{n^{1.5}}$.
8. Apply the Nth-Term Test to the series $\sum_{n=1}^{\infty} \frac{n+1}{n}$.
9. Use the Comparison Test to determine the convergence of $\sum_{n=1}^{\infty} \frac{1}{n^2+1}$.
10. Apply the Ratio Test to the series $\sum_{n=1}^{\infty} \frac{n!}{2^n}$.

Advanced Problems

11. Use the Integral Test to determine the convergence of $\sum_{n=2}^{\infty} \frac{1}{n \ln n}$.
12. Determine the convergence of the alternating series $\sum_{n=1}^{\infty} (-1)^{n+1} \frac{3}{n}$.
13. Use the Limit Comparison Test to analyze the convergence of $\sum_{n=1}^{\infty} \frac{5n+2}{3n^2+n}$.
14. Apply the Root Test to the series $\sum_{n=1}^{\infty} \left(\frac{n}{n+1}\right)^n$.
15. Determine whether the series $\sum_{n=1}^{\infty} \frac{(-1)^n}{\sqrt{n}}$ converges absolutely, conditionally, or diverges.

Summary

Sequences and series are powerful tools in calculus that help in understanding the behavior of functions and in solving complex problems involving sums and limits. Mastery of these concepts is essential for progressing in mathematical analysis and its applications.